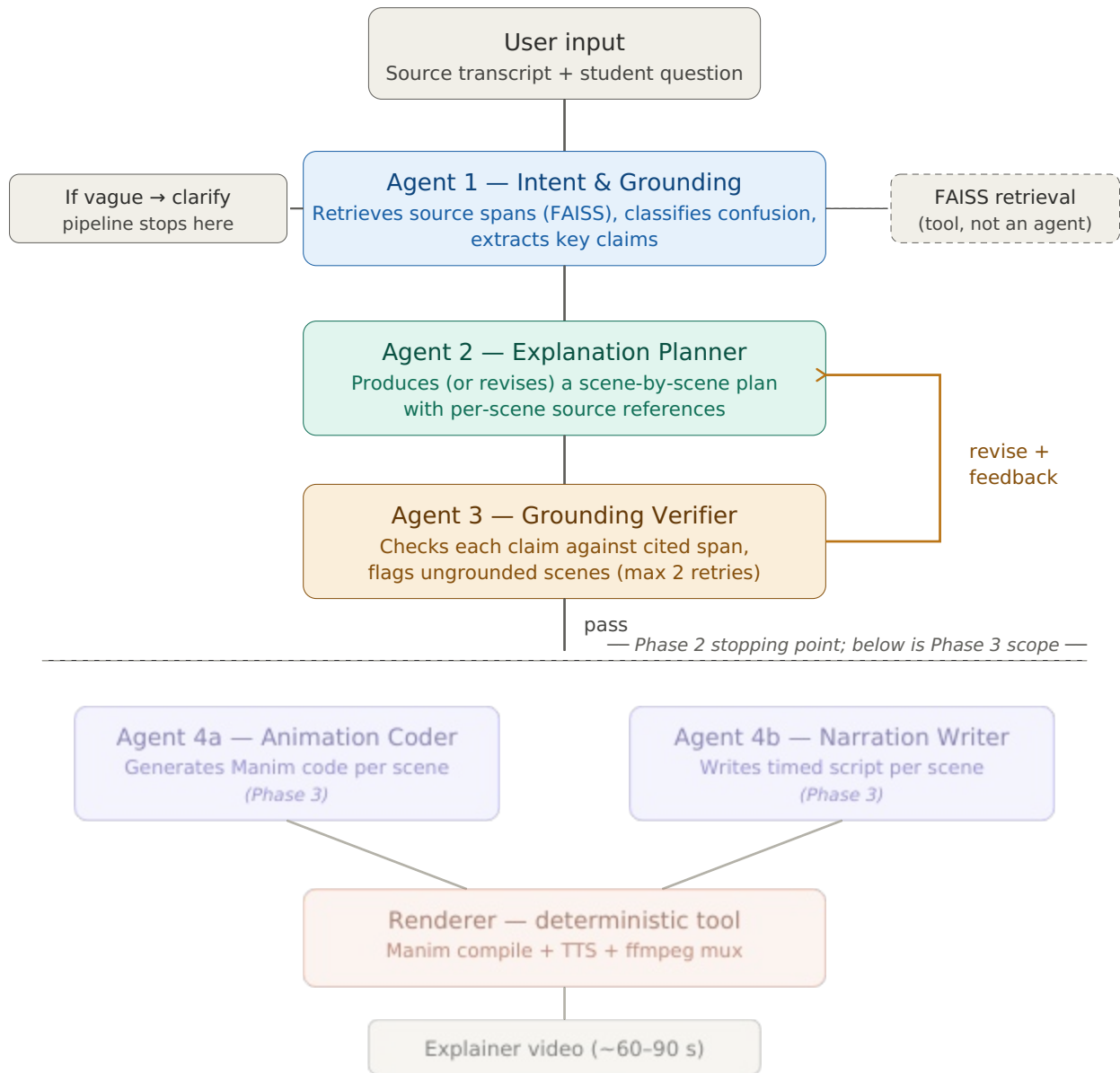


Aurua — Architecture Diagram

Jorge Urias · 94815 Agentic Technologies · CMU Heinz College · Phase 2



Legend

- ☐ Agent 1 — Intent & Grounding. Implemented. Uses FAISS retrieval as a tool; classifies confusion type; extracts key claims; can short-circuit on vague input.
- ☐ Agent 2 — Explanation Planner. Implemented. Produces structured scene plans; re-plans flagged scenes on verifier feedback.
- ☐ Agent 3 — Grounding Verifier. Implemented. Distinct system prompt; checks each claim against cited span; drives the revise loop (max 2 retries).
- ☐ Agents 4a / 4b. Phase 3. Animation code generation and narration writing run in parallel on the verified plan.
- ☐ Renderer — deterministic tool. Phase 3. No LLM. Composes Manim output, synthesized audio, and muxes the video with ffmpeg.
- ☐ External input/output and neutral utilities. User input, tool sidecars, short-circuit branches.

Coordination notes

The orange arrow from the Verifier back to the Planner is the feedback loop that makes the system genuinely agentic

rather than a prompt chain. The Planner receives per-scene verdicts with specific ungrounded-claim reasons and revises only the flagged scenes. Pass ends the Phase 2 pipeline. After two failed retries the system exits with status `verification_exhausted` rather than shipping ungrounded content. A vague-input short-circuit on Agent 1 avoids even entering the plan-verify loop when retrieval scores signal out-of-scope.